SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN COURSE CODE: CSA1102

COURSE OUTCOME COVERED

CO5: Analyze object-oriented software using quality assurance and usability testing Techniques (BL4,BL5)
Assessment Tool Weightage: 35%

Total Marks:35

QUESTIONS: 5

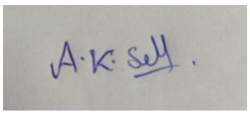
Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I A K Selva prabhu (192421224) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

Constraint-Based Problem Solving Assessment Rubric

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Annexure A

Constraint-Based Problem Solving

Q1. Scenario: Online Banking System

System Components:
Account Management, Fund Transfer, Bill Payment, Transaction History

Given Constraints:

- Transaction completion time < 2 seconds
- Multi-factor authentication required
- Support 5000+ concurrent users

Question:

Analyze the system using quality assurance techniques under the given constraints. Identify potential defects related to security, performance, and reliability. Propose suitable testing strategies and justify how they ensure system quality. **(BL4)**

Q2. Scenario: E-Commerce Shopping Application

System Features:
Product Search, Shopping Cart, Checkout, Payment Gateway

Usability Constraints:

- Checkout process within 3 steps
- Easy product navigation
- Responsive design for mobile and desktop devices

Question:

Apply usability testing techniques to evaluate the application. Identify usability challenges and recommend design improvements that satisfy the given constraints. Explain how the proposed changes improve customer satisfaction and efficiency. **(BL4)**

Q3. Scenario: Hospital Management System

Modules:
Patient Registration, Appointment Scheduling, Medical Records, Billing

QA Constraints:

- Patient data accuracy must be maintained
- System availability 24/7
- Simultaneous access by multiple departments

Question:

Use quality assurance practices to analyze the system. Identify critical test cases and discuss how the constraints influence testing priorities, defect prevention, and overall system reliability. **(BL5)**

Q4. Scenario: Smart Library Management System

Features:
Book Search, Book Reservation, Borrow/Return, Notifications

Usability Constraints:

- Search results displayed within 2 seconds
- Accessible interface for all age groups
- Minimal learning curve for first-time users

Question:

Conduct a usability evaluation using appropriate testing methods. Analyze how the given constraints impact interface design and recommend improvements to enhance usability and accessibility. **(BL5)**

Q5. Scenario: Airline Reservation System

Modules:
User Login, Flight Search, Seat Selection, Ticket Booking

QA & Usability Constraints:

- No booking duplication
- System must support peak holiday traffic

- Booking interface should be simple and error-free

Question:

Apply both quality assurance and usability testing techniques under the specified constraints. Identify major risks, propose suitable testing methods, and explain how the system can be optimized for reliability, performance, and user experience. **(BL4)**

Q1. Online Banking System – Constraint-Based Problem Solving

1. Quality Assurance Analysis

The online banking system contains Account Management, Fund Transfer, Bill Payment, and Transaction History. QA must ensure that the system is secure, fast, reliable, and capable of handling many users.

2. Potential Defects

Security defects:

- Weak password or MFA validation.
- Unauthorized access to accounts.
- Session hijacking and insecure sessions.
- Incorrect authorization during fund transfers.
- Sensitive banking data exposed in logs or responses.

Performance defects:

- Transaction completion taking more than 2 seconds.
- Slow response when many users access the system.
- Database bottlenecks during fund transfers.
- Server crashes under 5000+ concurrent users.

Reliability defects:

- Duplicate transactions.
- Money deducted without successful transfer.
- Transaction status not updated correctly.
- System failure during payment or transfer.

3. Testing Strategies

Testing Type	Purpose
Functional Testing	Verify account, transfer, bill payment and history functions
Security Testing	Test MFA, authorization, encryption and session security

Performance Testing	Verify transaction response is below 2 seconds
Load Testing	Test the system with 5000+ concurrent users
Stress Testing	Check behavior beyond normal capacity
Reliability Testing	Verify transactions remain correct during failures
Recovery Testing	Check recovery after server/database failure

4. Justification

These testing techniques identify security vulnerabilities, performance bottlenecks, transaction errors, and system failures. Load and performance testing verify the 2-second and 5000-user constraints, while security and reliability testing protect customer information and prevent financial errors.

Conclusion

A combination of security, performance, functional, load, and recovery testing ensures that the banking system is secure, responsive, reliable, and capable of supporting large numbers of users.

Q2. E-Commerce Shopping Application – Usability Testing

1. Usability Evaluation

The application includes Product Search, Shopping Cart, Checkout, and Payment Gateway. Usability testing checks whether users can find products easily and complete purchases quickly without confusion.

2. Usability Challenges

- Too many steps during checkout.
- Difficult product navigation.
- Poor search or filtering options.
- Confusing cart and payment information.
- Small buttons or text on mobile devices.
- Layout problems on different screen sizes.
- Error messages may not clearly explain the problem.

3. Testing Methods

User Testing:
Ask real users to search for products, add items to the cart, and complete checkout.

Task-Based Testing:
Give users tasks such as *"Find a laptop and purchase it."* Measure completion time and errors.

A/B Testing:
Compare two different checkout or navigation designs.

Responsive Testing:
Test the application on mobile, tablet, and desktop screens.

Accessibility Testing:
Check text readability, keyboard navigation, contrast, and screen-reader support.

4. Recommended Improvements

- Keep checkout within 3 simple steps.
- Provide clear categories and search filters.
- Add breadcrumbs for easy navigation.
- Use large, clearly labeled buttons.
- Display cart summary before payment.
- Provide clear error and validation messages.
- Use responsive layouts for different devices.
- Provide consistent navigation across pages.

5. Benefits

These improvements reduce user confusion and the time required to complete purchases. A simple checkout reduces cart abandonment, while responsive and accessible design improves customer satisfaction.

Conclusion

Usability testing helps create an e-commerce application that is simple, fast, accessible, and convenient, resulting in better customer experience and higher purchasing efficiency.

Q3. Hospital Management System – Quality Assurance

1. QA Analysis

The Hospital Management System manages sensitive patient information through Patient Registration, Appointment Scheduling, Medical Records, and Billing. QA must prioritize accuracy, availability, security, and data consistency.

2. Critical Test Cases

Patient Registration

- Verify correct patient details are stored.

- Check duplicate patient registration.
- Validate mandatory fields.
- Verify patient ID generation.

Appointment Scheduling

- Check doctor availability.
- Prevent double-booking.
- Verify appointment date and time.
- Test cancellation and rescheduling.

Medical Records

- Verify correct patient records are displayed.
- Prevent unauthorized access.
- Check data update and retrieval.
- Ensure records are not accidentally deleted.

Billing

- Verify bill calculations.
- Check insurance/discount calculations.
- Verify payment status.
- Prevent duplicate billing.

3. Testing Priorities

Constraint	Testing Priority
Patient data accuracy	Data validation & functional testing
24/7 availability	Availability & recovery testing
Multiple departments	Concurrency & integration testing
Sensitive records	Security testing
Billing accuracy	Calculation & transaction testing

4. Defect Prevention

- Use input validation.
- Apply database constraints.
- Use role-based access control.
- Perform code reviews.
- Maintain automated regression tests.
- Use database backup and recovery mechanisms.
- Validate data before storing it.

5. Reliability

Concurrency testing ensures that doctors, nurses, billing staff, and administrators can access the system simultaneously without data conflicts. Availability and recovery testing ensure that the system continues operating and can recover quickly from failures.

Conclusion

QA practices help maintain accurate patient data, continuous availability, secure records, and reliable multi-department operations, making the hospital system dependable.

Q4. Smart Library Management System – Usability Evaluation

1. Usability Evaluation

The Smart Library system provides Book Search, Reservation, Borrow/Return, and Notifications. The system should be fast, accessible, and easy to learn, especially for first-time users.

2. Usability Testing Methods

User Testing:
Observe students, teachers, and other users while they search and reserve books.

Task-Based Testing:
Ask users to perform tasks such as *"Find a book and reserve it."*

Accessibility Testing:
Check font size, color contrast, keyboard navigation, and screen-reader compatibility.

Performance Testing:
Verify that search results appear within 2 seconds.

First-Time User Testing:
Measure how easily a new user can complete common tasks without assistance.

3. Usability Challenges

- Complicated search interface.
- Small or unclear text.
- Poor contrast.
- Too many menu options.
- Confusing reservation process.
- Lack of clear feedback after borrowing or reservation.
- Difficult navigation for older users.

4. Recommended Improvements

- Provide a simple search bar on the home page.
- Add filters such as author, category, and availability.
- Use readable fonts and appropriate font sizes.
- Provide high-contrast interface options.
- Use clear icons with text labels.
- Keep reservation steps short.
- Display clear success/error messages.
- Provide simple navigation and help instructions.

5. Impact

A fast search system saves users' time. Accessible design allows people of different ages and abilities to use the system comfortably. Simple navigation reduces the learning curve and allows first-time users to complete tasks independently.

Conclusion

Usability and accessibility testing ensure that the library system is fast, simple, inclusive, and easy to use.

Q5. Airline Reservation System – QA & Usability Testing

1. QA Analysis

The airline reservation system includes User Login, Flight Search, Seat Selection, and Ticket Booking. The main quality requirements are no duplicate bookings, high performance during peak traffic, and a simple booking interface.

2. Major Risks

- Two users booking the same seat.
- Duplicate ticket generation.
- Payment completed but ticket not generated.
- Flight search becoming slow during holidays.
- Server failure during booking.
- Incorrect passenger or flight information.
- Users selecting incorrect dates or seats.

3. Testing Methods

Testing Method	Purpose

Functional Testing	Verify login, search, seat selection and booking
Security Testing	Protect user and payment information
Concurrency Testing	Prevent two users from booking the same seat
Load Testing	Test peak holiday traffic
Stress Testing	Check behavior beyond expected traffic
Recovery Testing	Verify recovery after system failures
Usability Testing	Check whether booking is simple
Compatibility Testing	Test different browsers and devices
Regression Testing	Ensure new changes do not break existing functions

4. Usability Improvements

- Use a simple step-by-step booking process.
- Provide clear flight search filters.
- Display available seats visually.
- Highlight selected seats clearly.
- Show a booking summary before confirmation.
- Validate passenger details before payment.
- Provide clear error messages.
- Display confirmation immediately after successful booking.

5. Reliability Improvements

- Use database transactions for booking.
- Lock a selected seat temporarily during payment.
- Prevent duplicate booking using unique booking/seat constraints.
- Use automatic backups.
- Implement server redundancy and failover.
- Maintain transaction logs for recovery.

6. Performance Optimization

Load testing should simulate peak holiday traffic. Caching can speed up flight searches, while database optimization and scalable servers can handle large numbers of users.

Conclusion

Combining QA and usability testing ensures that the airline reservation system is reliable, fast, secure, and easy to use. Preventing duplicate bookings and testing peak traffic are the highest priorities because they directly affect customer trust and system reliability.